Financial-document triage across public sources

This test does not resolve a macro-F1 advantage for adapted Laya over the selected linear baseline.

The primary validation-selected contrast is Laya LoRA versus Word TF-IDF logistic. Test macro F1 is 0.712 versus 0.701; paired delta 95% interval [-0.013, 0.036]. These are agreement scores against Luna labels.

Independent audit: Deferred for this version by agreement; 0 resolved and 0 unresolved documents. No claim of analyst usefulness or investment profitability follows from teacher agreement alone.

Findings and engineering implications

The primary systems have 177 and 187 default critical misses, respectively, among 1292 teacher review-worthy documents. Their warm mean latencies are 267.8 and 4.52 ms on the common selection sample (59.2 times the latency for the decision model). The small observed F1 difference and its interval make a strong lexical baseline essential to the comparison.

Increasing Laya LoRA from 1000 to 4000 training documents raises observed F1 by 0.071; paired 95% interval [0.049, 0.093]. This supports sample efficiency within this fixed recipe and cohort. Changing subset composition and additional selection/calibration labels prevent interpreting it as a universal minimum-data requirement.

FinBERT LoRA has observed F1 0.642 and 306 critical misses, compared with 0.666 and 236 for frozen FinBERT/logistic regression. These are point estimates for the declared recipes, not an exhaustive fine-tuning comparison. Gains over a fresh random triage head alone do not establish that adapters outperform a trained frozen-feature classifier.

The completed head-context ablation does not show that wider context improves every metric. F1 and critical misses need to be read together under the fixed pooling rule. These supplementary comparisons do not justify selecting a new window size on the test cohort.

The primary model's calibration-selected 1%/5% discard rules remove 5.39%/14.72% of the test workload while missing 1.32%/6.58% of teacher review-worthy documents. Calibration targets are empirical constraints, not deployment bounds. The observed rates and support matter more than the target names; confidence alone cannot certify safe automatic discarding.

This bounded study establishes an inspectable end-to-end comparison. Analyst usefulness and commercial-inbox generalization still require independent labels and representative source coverage. Any further recipe or policy selection needs fresh held-out data; the current test is closed to tuning.

Task and evaluation

The primary contrast and main trained-model tables use 4000 training documents per method. Unadapted references use no task training; larger imported curves are supplementary. Generative LLM adaptation, larger LoRA runs and independent human review are deferred by the user. This is an end-to-end teacher-agreement study, not expert validation of materiality.

Classify supplied source content as routine, review_worthy or unclear for an analyst inbox. No issuer history, portfolio context, future returns or market reaction is supplied. Unclear routes to review. Macro F1 averages all three declared labels, including absent classes in small slices. The critical error is a review-worthy document predicted routine.

The research question is whether supervised decision-model adaptation and larger contiguous context improve triage agreement, critical misses and throughput relative to lexical and encoder baselines. FinBERT's published sentiment results do not establish triage quality; BGE is an embedding model. ModernBERT supplies a backbone-family system comparison; classifier/decision heads and source-window budgets differ. Primary sources and distinctions are documented in literature.md.

Train ends 2026-03-31; selection/calibration use April-June; test uses July-September. Event and verified near-duplicate groups cannot cross boundaries. Old experimental bodies and near duplicates are excluded from fresh validation/test. Candidate retrieval is approximate and can miss duplicates. Every core recipe, checkpoint,

calibration and discard threshold was frozen before test inference.

Pretrained model	Pinned Hub revision
BAAI/bge-small-en-v1.5	5c38ec7c405ec4b44b94cc5a9bb96e735b38267a
ProsusAI/finbert	4556d13015211d73dccd3fdd39d39232506f3e43
answerdotai/ModernBERT-large	45bb4654a4d5aaff24dd11d4781fa46d39bf8c13
convaiinnovations/laya	55cf4c4ebb4ebe31b2550e8bdf3bd21b99753851

Source and label coverage

Input	Raw documents	Family / provider	Document dates
8k.jsonl	30000	8k / sec	2019-01-02 to 2026-09-30
releases.jsonl	30000	releases / sec	2019-01-02 to 2026-09-30
6k.jsonl	25000	6k / sec	2019-01-02 to 2026-09-30
fed.jsonl	1826	fed / fed	2019-01-08 to 2026-09-30
ecb.jsonl	722	ecb / ecb	2019-01-10 to 2026-09-29
news.jsonl	201	news / wikinews	2019-01-05 to 2026-04-15
govuk.jsonl	1257	news / govuk	2019-01-01 to 2026-09-30
dataset.json (legacy)	10000	8k / sec	2025-10-01 to 2026-09-30

Raw acquisition counts precede duplicate quarantine, sampling and labeling. Legacy inputs were previously experimented on and are eligible only for training. SEC dates are filing dates, including release exhibits, and may be later than publication elsewhere. Other sources use original article dates or publication metadata; temporal splits use these recorded document dates at day precision.

Legacy text formatter	Primary filings reconstructed	Exact old-body matches
edgartools 5.59.1	30000	588

Offline reconstruction of retained primary HTML links exact old-body hashes across text-extraction formats. These links also quarantine related release exhibits from fresh evaluation. Matching candidates are not additional exclusion counts; their groups enter the exclusion rules below. Canonical model inputs are unchanged.

Legacy event formatter	Old bodies covered	Verified events	Matching source families
edgartools 5.59.1	10000	10000	8k: 610, releases: 593

Every old body is reproduced exactly from preserved submissions or retained index/primary responses. The resulting filing-event links also quarantine release-only candidates whose parent filing is absent from the new sample. Matching source rows enter the group exclusion rules and are not additional exclusion counts. Evidence hashes and reconstruction methods are recorded with the corpus.

Family	Exclusion reason	Documents
8k	cross boundary group	555
8k	previously experimented group	5132
6k	cross boundary group	138
releases	cross boundary group	10
releases	previously experimented group	294

Partition	Available after exclusions	Selected before labeling
train	87793	25000
validation	2580	1200
test	2504	1800

Selection applies the recorded split ceilings and train/validation/test 8-K share cap; these counts are distinct from labeled support below.

Family	Train	Selection	Calibration	Test
6k	7134	243	111	561
8k	7068	246	108	561
ecb	666	17	12	27
fed	1712	44	15	55
news	1373	38	12	35
releases	7047	249	105	561

Source shortfalls are not filled by additional 8-Ks. These institutional/public-news cohorts are not a representative commercial inbox. Publication dates and snapshots cannot establish the exact text available at the historical date; pretraining exposure remains possible. Rights and collection methods are documented in sources.md.

Split	Documents	routine	review_worthy	unclear
train	25000	5569	16674	2757
selection	837	192	552	93
calibration	363	75	252	36
test	1800	301	1292	207

Provider	Train	Selection	Calibration	Test
ecb	666	17	12	27
fed	1712	44	15	55
govuk	1173	37	12	35
sec	21249	738	324	1683
wikinews	200	1	0	0

The news family can contain official financial announcements as well as Wikinews. Provider coverage is reported separately; official announcements do not establish commercial-journalism generalization.

Label queue	Queued	Labeled	Unlabeled	Retained	8-K cap excluded
train	25000	25000	0	25000	0
validation	1200	1200	0	1200	0
test	1800	1800	0	1800	0

Validation/test labeling must be complete. A budget-limited training queue can retain unlabeled rows; family/error counts remain in the manifest. The 40% primary 8-K cap is checked again on retained labels; cap exclusions remain in the input queue and project spending. Attained sample sizes, rather than collection targets, define the learning curves.

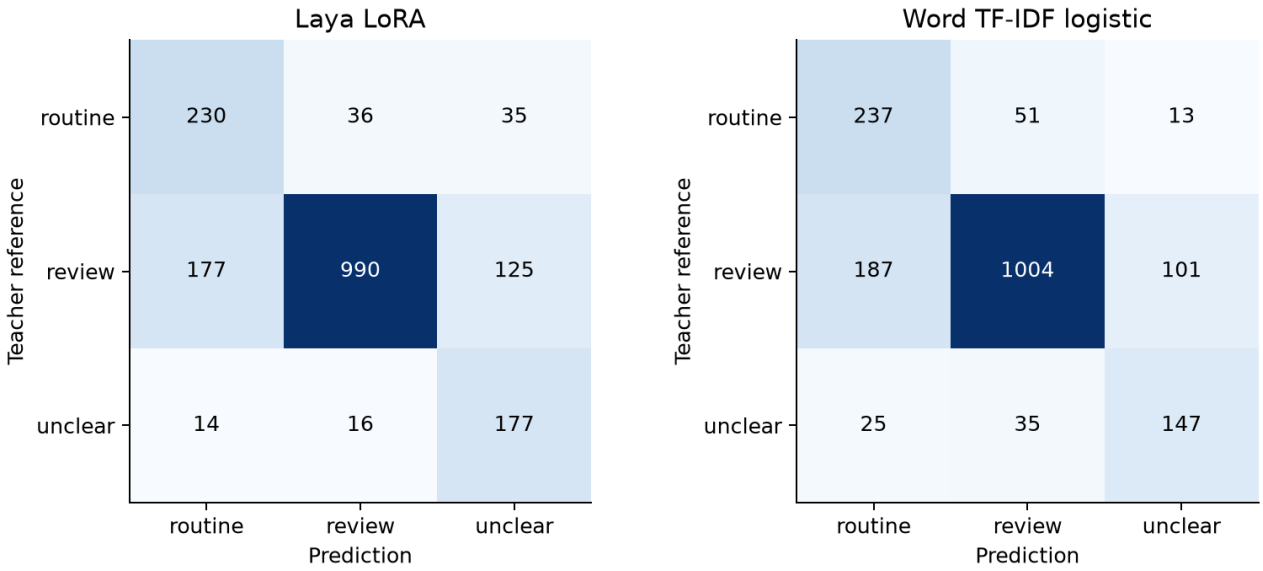
Label queue	Successful teacher labels	Recorded USD	Cost unavailable
train	25000	4.3940	0
validation	1200	0.1730	0
test	1800	0.3178	0

These costs cover successful teacher annotations retained in the frozen benchmark, using recorded API usage and prices. Missing costs are counted separately. The shared $5 project ledger also covers pilots, failed or unresolved requests and labels excluded from the benchmark; these retained-label totals are not the entire project bill. Human-review time is excluded.

Model comparisons

Model	Train	Macro F1	95% CI	Misses	Miss rate	Brier
Laya LoRA	4000	0.712	[0.686, 0.735]	177	0.137	0.300
Word TF-IDF logistic	4000	0.701	[0.674, 0.726]	187	0.145	0.304
Majority class	4000	0.279	[0.274, 0.284]	0	0.000	0.446
Word TF-IDF svm	4000	0.683	[0.656, 0.709]	176	0.136	0.317
Char TF-IDF logistic	4000	0.692	[0.664, 0.716]	230	0.178	0.339
Char TF-IDF svm	4000	0.680	[0.652, 0.705]	207	0.160	0.339
Combined TF-IDF logistic	4000	0.702	[0.675, 0.727]	183	0.142	0.308
Combined TF-IDF svm	4000	0.698	[0.671, 0.723]	194	0.150	0.314
FinBERT logistic	4000	0.666	[0.641, 0.691]	236	0.183	0.368
BGE-small logistic	4000	0.639	[0.613, 0.664]	238	0.184	0.390
Laya SDK	0	0.363	[0.348, 0.379]	321	0.248	0.488
Laya head	4000	0.545	[0.519, 0.570]	231	0.179	0.471
ModernBERT-large LoRA	4000	0.689	[0.664, 0.713]	267	0.207	0.344
FinBERT LoRA	4000	0.642	[0.616, 0.666]	306	0.237	0.428

Misses use the default argmax routine decision; the denominator is review-worthy support. Brier sums squared errors over three classes (0-2). Full artifacts include class precision/recall/F1, confusion matrices, accuracy, log loss, ten-bin ECE, source slices and reference provenance. Intervals resample whole related groups 2000 times with shared draws; they condition on the single trained seed and have no multiplicity correction.



Confusion matrices show the critical review-to-routine cell

Model	Class	Precision	Recall	F1	Support
Laya LoRA	routine	0.546	0.764	0.637	301
Laya LoRA	review_worthy	0.950	0.766	0.848	1292
Laya LoRA	unclear	0.525	0.855	0.651	207
Word TF-IDF logistic	routine	0.528	0.787	0.632	301
Word TF-IDF logistic	review_worthy	0.921	0.777	0.843	1292
Word TF-IDF logistic	unclear	0.563	0.710	0.628	207

Source slices and uncertainty

Model	Family	Documents	Macro F1	Review support	Misses
Laya LoRA	6k	561	0.740	297	40
Laya LoRA	8k	561	0.607	413	70
Laya LoRA	ecb	27	0.500	18	3
Laya LoRA	fed	55	0.483	40	9
Laya LoRA	news	35	0.563	25	5
Laya LoRA	releases	561	0.530	499	50
Word TF-IDF logistic	6k	561	0.684	297	45
Word TF-IDF logistic	8k	561	0.631	413	73
Word TF-IDF logistic	ecb	27	0.564	18	4
Word TF-IDF logistic	fed	55	0.544	40	8
Word TF-IDF logistic	news	35	0.582	25	4
Word TF-IDF logistic	releases	561	0.510	499	53

Model	Provider	Documents	Macro F1	Review support	Misses
Laya LoRA	ecb	27	0.500	18	3
Laya LoRA	fed	55	0.483	40	9

Model	Provider	Documents	Macro F1	Review support	Misses
Laya LoRA	govuk	35	0.563	25	5
Laya LoRA	sec	1683	0.712	1209	160
Word TF-IDF logistic	ecb	27	0.564	18	4
Word TF-IDF logistic	fed	55	0.544	40	8
Word TF-IDF logistic	govuk	35	0.582	25	4
Word TF-IDF logistic	sec	1683	0.693	1209	171

Model	Log loss	ECE (10 bins)	Median uncertainty
Laya LoRA	0.501	0.034	0.147
Word TF-IDF logistic	0.502	0.028	0.158
Majority class	0.792	0.024	0.306
Word TF-IDF svm	0.535	0.038	0.155
Char TF-IDF logistic	0.560	0.024	0.200
Char TF-IDF svm	0.567	0.027	0.193
Combined TF-IDF logistic	0.509	0.021	0.164
Combined TF-IDF svm	0.523	0.033	0.163
FinBERT logistic	0.605	0.015	0.266
BGE-small logistic	0.647	0.038	0.286
Laya SDK	0.838	0.115	0.328
Laya head	0.781	0.041	0.359
ModernBERT-large LoRA	0.560	0.018	0.207
FinBERT LoRA	0.692	0.042	0.280

Model	Raw Brier	Cal. Brier	Raw ECE	Cal. ECE
Laya LoRA	0.303	0.300	0.049	0.034
Word TF-IDF logistic	0.313	0.304	0.067	0.028
Majority class	0.564	0.446	0.282	0.024
Word TF-IDF svm	0.321	0.317	0.056	0.038
Char TF-IDF logistic	0.366	0.339	0.110	0.024
Char TF-IDF svm	0.338	0.339	0.029	0.027
Combined TF-IDF logistic	0.338	0.308	0.111	0.021
Combined TF-IDF svm	0.328	0.314	0.093	0.033
FinBERT logistic	0.368	0.368	0.017	0.015
BGE-small logistic	0.397	0.390	0.067	0.038
Laya SDK	0.486	0.488	0.110	0.115
Laya head	0.472	0.471	0.072	0.041
ModernBERT-large LoRA	0.348	0.344	0.052	0.018
FinBERT LoRA	0.452	0.428	0.107	0.042

Raw and calibrated scores use the same test documents. A scalar temperature is fitted on separate calibration groups; it changes confidence and discard behavior while preserving argmax labels. Its use follows Guo et al.; it does not guarantee calibration on this cohort.

Uncertainty is 1 minus the largest calibrated class probability. It is model confidence, not a probability of investment materiality; teacher self-reported uncertainty is retained separately. Class support must be read beside small-slice macro F1. When no misses are observed, bootstrap resampling cannot reveal unseen errors.

Calibrated uncertainty: Laya LoRA.

Uncertainty	Documents	Accuracy	Review support	Misses	Miss rate
[0.00, 0.10)	696	0.954	512	21	0.041
[0.10, 0.20)	386	0.824	283	34	0.120
[0.20, 0.40)	460	0.635	321	65	0.202
[0.40, 1.00]	258	0.477	176	57	0.324

Calibrated uncertainty: Word TF-IDF logistic.

Uncertainty	Documents	Accuracy	Review support	Misses	Miss rate
[0.00, 0.10)	706	0.963	572	18	0.031
[0.10, 0.20)	306	0.794	199	37	0.186
[0.20, 0.40)	475	0.663	314	69	0.220
[0.40, 1.00]	313	0.479	207	63	0.304

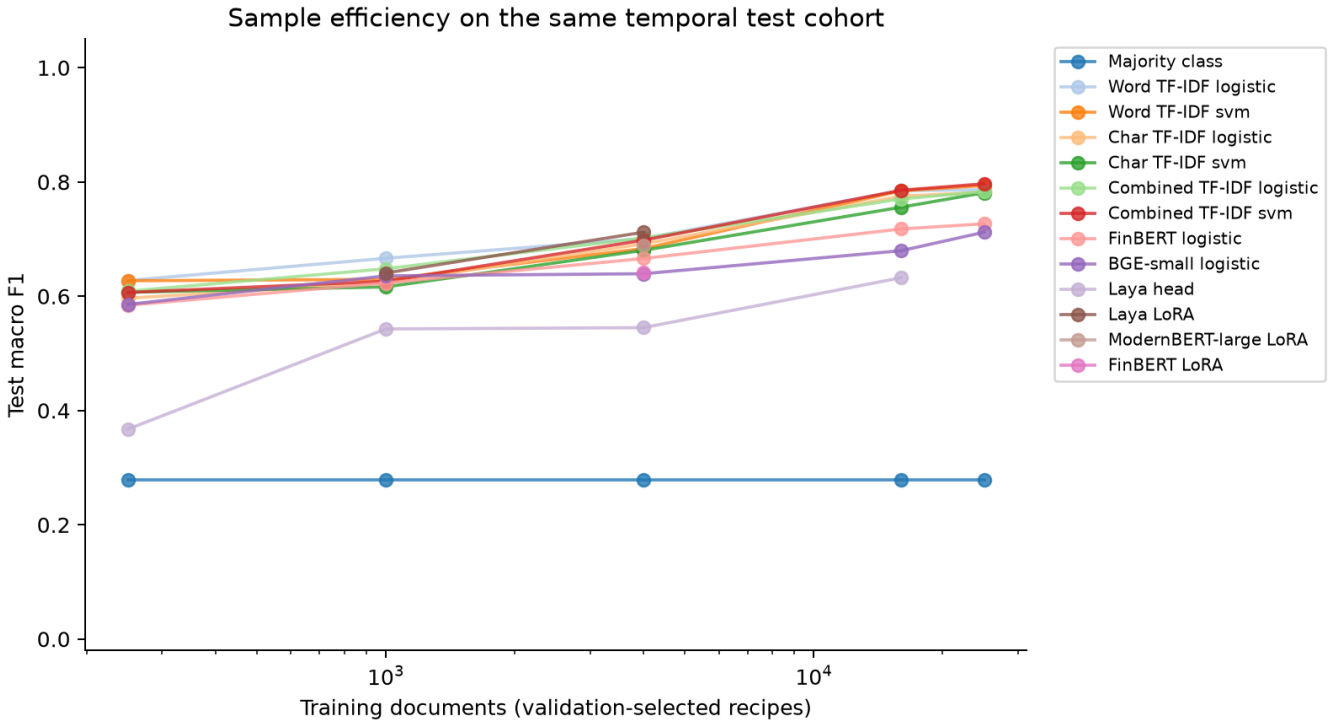
Bands are fixed before testing and descriptive, not selected discard thresholds. Boundary values enter the higher band; the final band includes its upper bound. Miss rates divide default review-to-routine misses by review-worthy support within each band. Empty bands and bands without review support have undefined rates, shown as dashes. Three-class uncertainty cannot exceed 2/3; the final band's upper bound covers all valid values.

Discard policies

Model	Target	Workload reduction	Misses	Support	Miss rate	Contamination
Laya LoRA	0.01	0.054	17	1292	0.013	0.196
Laya LoRA	0.05	0.147	85	1292	0.066	0.336
Word TF-IDF logistic	0.01	0.067	16	1292	0.012	0.157
Word TF-IDF logistic	0.05	0.134	65	1292	0.050	0.306
Majority class	0.01	0.000	0	1292	0.000	-
Majority class	0.05	0.000	0	1292	0.000	-
Word TF-IDF svm	0.01	0.052	10	1292	0.008	0.129
Word TF-IDF svm	0.05	0.119	49	1292	0.038	0.262
Char TF-IDF logistic	0.01	0.033	10	1292	0.008	0.169
Char TF-IDF logistic	0.05	0.113	59	1292	0.046	0.325
Char TF-IDF svm	0.01	0.037	12	1292	0.009	0.197
Char TF-IDF svm	0.05	0.090	51	1292	0.039	0.346
Combined TF-IDF logistic	0.01	0.061	15	1292	0.012	0.156
Combined TF-IDF logistic	0.05	0.115	51	1292	0.039	0.271
Combined TF-IDF svm	0.01	0.059	16	1292	0.012	0.170
Combined TF-IDF svm	0.05	0.116	55	1292	0.043	0.292
FinBERT logistic	0.01	0.039	16	1292	0.012	0.225
FinBERT logistic	0.05	0.121	79	1292	0.061	0.382
BGE-small logistic	0.01	0.046	34	1292	0.026	0.410
BGE-small logistic	0.05	0.088	60	1292	0.046	0.377
Laya SDK	0.01	0.006	7	1292	0.005	0.800
Laya SDK	0.05	0.074	69	1292	0.053	0.744
Laya head	0.01	0.017	18	1292	0.014	0.613
Laya head	0.05	0.042	48	1292	0.037	0.671
ModernBERT-large LoRA	0.01	0.032	12	1292	0.009	0.207
ModernBERT-large LoRA	0.05	0.119	68	1292	0.053	0.322
FinBERT LoRA	0.01	0.032	18	1292	0.014	0.316
FinBERT LoRA	0.05	0.106	66	1292	0.051	0.363

Thresholds maximize discarded routine predictions subject to empirical calibration miss targets of 1%/5%. Predicted review-worthy/unclear always escalate. Sparse support and zero observed misses do not establish population guarantees. Test rates, rather than calibration targets, describe the observed deployment tradeoff.

Adaptation, sample efficiency and context



Nested learning curves on the same held-out cohort

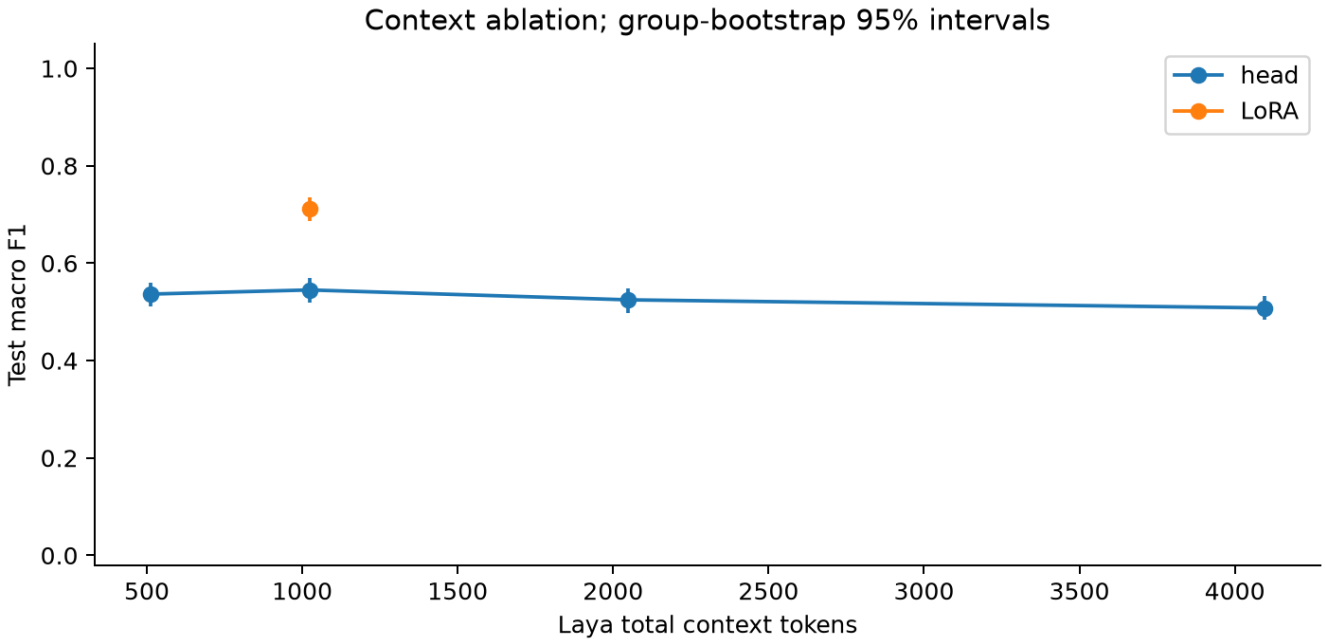
Training sizes are nested 250/1000/4000/16000/all where attainable. Each neural run restarts from pinned pretrained weights. Deterministic family/label round-robin sampling changes class/source proportions as strata exhaust; the allocation must be considered when interpreting sample efficiency. One training seed is used. The bounded study adds one predeclared 4000-document LoRA recipe each for Laya, ModernBERT and FinBERT, without a new hyperparameter sweep. Imported head recipes were previously selected at 1000/4000; their tuning labels are additional to the stated training size.

Training documents	Independent groups	routine	review_worthy	unclear
250	250	99	100	51
1000	999	403	404	193
4000	3991	1648	1720	632
16000	15824	5535	7987	2478
25000	24685	5569	16674	2757

Family	250	1000	4000	16000	25000
6k	51	204	861	5463	7134
8k	51	202	861	4136	7068
ecb	34	134	502	666	666
fed	49	191	631	1712	1712
news	32	134	572	1373	1373
releases	33	135	573	2650	7047

Provider	250	1000	4000	16000	25000
ecb	34	134	502	666	666
fed	49	191	631	1712	1712
govuk	27	109	481	1173	1173
sec	135	541	2295	12249	21249
wikinews	5	25	91	200	200

Counts use the exact frozen training subsets shared by all models. Independent groups may contain multiple related documents. These tables describe training allocation; recipe-screening and validation labels are additional.



Context ablations retain the same whole documents

Laya uses fixed uniform whole-document pooling and 50% overlapping windows; the native SDK reference uses most-confident-window aggregation. CE/Brier objectives are document-level; no chunk labels are invented. Matched pre-update controls separate supervised adaptation from aggregation and k-bit precision changes. FinBERT/BGE/ModernBERT use fresh triage heads; FinBERT's sentiment head is not mapped to triage labels.

Laya's adapted wrapper averages native three-choice window logits. Literal reserved mask tokens are replaced with spaces before formatting. FinBERT/ModernBERT use non-overlapping source windows, average unpadded token states within each window, then uniformly average window vectors before a new LayerNorm/dropout/linear classifier. Frozen FinBERT/BGE baselines normalize the mean of all window vectors for logistic regression; BGE uses CLS rather than mean token states. Lexical baselines process full bodies. These systems differ in tokenization, window overlap and heads, so their differences do not isolate encoder architecture alone. LoRA updates Wqkv in Laya/ModernBERT and query/value projections in FinBERT: rank 8, alpha 16, adapter dropout 0.05.

Adapted model	Train	Matched-zero F1	Adapted F1	Paired gain 95% CI
Laya LoRA	4000	0.364	0.712	[0.317, 0.377]
Laya head	4000	0.364	0.545	[0.151, 0.211]
ModernBERT-large LoRA	4000	0.096	0.689	[0.569, 0.617]
FinBERT LoRA	4000	0.097	0.642	[0.521, 0.568]

Model	Training increment	F1 gain	Paired gain 95% CI
Majority class	250 to 1000	0.000	[0.000, 0.000]
Majority class	1000 to 4000	0.000	[0.000, 0.000]
Majority class	4000 to 16000	0.000	[0.000, 0.000]
Majority class	16000 to 25000	0.000	[0.000, 0.000]
Word TF-IDF logistic	250 to 1000	0.039	[0.022, 0.056]
Word TF-IDF logistic	1000 to 4000	0.035	[0.014, 0.056]
Word TF-IDF logistic	4000 to 16000	0.084	[0.061, 0.109]
Word TF-IDF logistic	16000 to 25000	0.002	[-0.013, 0.016]
Word TF-IDF svm	250 to 1000	0.002	[-0.012, 0.017]
Word TF-IDF svm	1000 to 4000	0.054	[0.030, 0.077]
Word TF-IDF svm	4000 to 16000	0.101	[0.079, 0.126]
Word TF-IDF svm	16000 to 25000	0.009	[-0.003, 0.021]
Char TF-IDF logistic	250 to 1000	0.027	[0.013, 0.041]
Char TF-IDF logistic	1000 to 4000	0.068	[0.044, 0.092]
Char TF-IDF logistic	4000 to 16000	0.084	[0.060, 0.109]
Char TF-IDF logistic	16000 to 25000	0.007	[-0.010, 0.024]
Char TF-IDF svm	250 to 1000	0.009	[-0.015, 0.032]
Char TF-IDF svm	1000 to 4000	0.064	[0.034, 0.094]
Char TF-IDF svm	4000 to 16000	0.076	[0.053, 0.100]
Char TF-IDF svm	16000 to 25000	0.025	[0.006, 0.045]
Combined TF-IDF logistic	250 to 1000	0.039	[0.023, 0.056]
Combined TF-IDF logistic	1000 to 4000	0.054	[0.031, 0.077]
Combined TF-IDF logistic	4000 to 16000	0.068	[0.044, 0.093]
Combined TF-IDF logistic	16000 to 25000	0.013	[-0.005, 0.030]
Combined TF-IDF svm	250 to 1000	0.019	[0.004, 0.035]
Combined TF-IDF svm	1000 to 4000	0.072	[0.051, 0.094]
Combined TF-IDF svm	4000 to 16000	0.087	[0.065, 0.112]
Combined TF-IDF svm	16000 to 25000	0.011	[-0.002, 0.024]
FinBERT logistic	250 to 1000	0.039	[0.018, 0.060]
FinBERT logistic	1000 to 4000	0.043	[0.024, 0.063]
FinBERT logistic	4000 to 16000	0.052	[0.030, 0.075]
FinBERT logistic	16000 to 25000	0.009	[-0.014, 0.030]
BGE-small logistic	250 to 1000	0.050	[0.030, 0.070]
BGE-small logistic	1000 to 4000	0.004	[-0.016, 0.021]
BGE-small logistic	4000 to 16000	0.040	[0.019, 0.060]
BGE-small logistic	16000 to 25000	0.032	[0.011, 0.053]
Laya head	250 to 1000	0.175	[0.148, 0.202]
Laya head	1000 to 4000	0.002	[-0.026, 0.028]

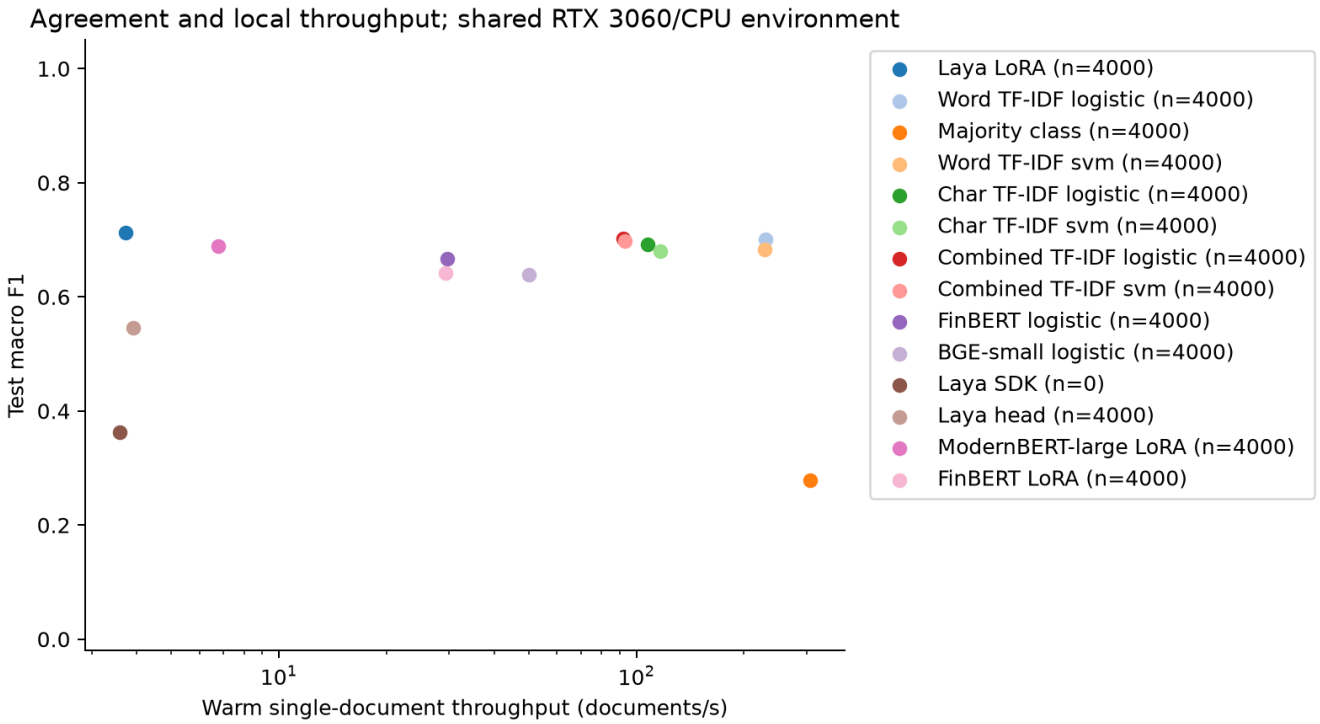
Model	Training increment	F1 gain	Paired gain 95% CI
Laya head	4000 to 16000	0.087	[0.064, 0.112]
Laya LoRA	1000 to 4000	0.071	[0.049, 0.093]

Model	Context tokens	F1 gain	Paired gain 95% CI
Laya head	512 to 1024	0.009	[-0.014, 0.031]
Laya head	512 to 2048	-0.012	[-0.036, 0.012]
Laya head	512 to 4096	-0.028	[-0.052, -0.006]

Wider Laya windows change contiguous context and the number of votes under the same pooling rule. All encoded source tokens after tokenizer/reserved-token normalization remain covered at each context size; the difference is not previously omitted text.

These paired differences describe this fixed-seed cohort. Additional tuning labels and multiple comparisons limit claims about minimum data requirements or statistical significance.

Latency and cost



Warm throughput measured locally

Model	P50 s	P95 s	Batch 1 docs/s	Batch 8 docs/s	Peak VRAM GiB
Laya LoRA	0.144	1.128	3.737	3.737	2.11
Word TF-IDF logistic	0.004	0.008	229.703	678.769	0.00
Majority class	0.003	0.003	305.830	2918.410	0.00
Word TF-IDF svm	0.004	0.008	228.751	661.454	0.00
Char TF-IDF logistic	0.006	0.023	107.762	180.121	0.00
Char TF-IDF svm	0.006	0.023	116.543	179.075	0.00
Combined TF-IDF logistic	0.007	0.029	92.004	143.477	0.00

Model	P50 s	P95 s	Batch 1 docs/s	Batch 8 docs/s	Peak VRAM GiB
Combined TF-IDF svm	0.007	0.029	92.816	141.649	0.00
FinBERT logistic	0.021	0.109	29.642	33.083	0.51
BGE-small logistic	0.015	0.054	50.057	61.110	0.18
Laya SDK	0.149	1.086	3.595	3.584	4.07
Laya head	0.137	1.065	3.928	3.923	2.09
ModernBERT-large LoRA	0.072	0.525	6.771	6.798	1.81
FinBERT LoRA	0.020	0.118	29.323	29.558	0.52

Warm timings include preprocessing and all model windows, exclude loading, and use a fixed 128-document selection sample on a shared machine. Sklearn batches are native; neural/SDK/prompt document batches are serial, with window batching where supported. Local API charges are zero; hardware, electricity and collection are additional.

Model	Fit elapsed s	Trainable neural M	Training VRAM GiB	Original weights GiB	Saved checkpoint MiB
Laya LoRA	13962.5	27.166	3.18	-	311.15
Word TF-IDF logistic	38.6	-	0.00	-	6.22
Majority class	0.1	-	0.00	-	0.00
Word TF-IDF svm	52.0	-	0.00	-	6.22
Char TF-IDF logistic	109.7	-	0.00	-	5.57
Char TF-IDF svm	270.5	-	0.00	-	5.57
Combined TF-IDF logistic	139.0	-	0.00	-	11.79
Combined TF-IDF svm	354.4	-	0.00	-	11.79
FinBERT logistic	32.7	-	0.00	0.41	0.03
BGE-small logistic	12.4	-	0.00	0.12	0.01
Laya SDK	0.0	-	0.00	-	0.00
Laya head	7160.4	26.248	2.35	-	300.59
ModernBERT-large LoRA	7813.7	0.923	2.88	1.47	10.79
FinBERT LoRA	1958.6	0.299	0.78	0.41	3.65

Fit elapsed includes training/grid selection and recorded validation work; cached features can reduce elapsed time. Neural trainable counts include heads and adapters. Equal LoRA rank does not imply equal updated parameter budgets: Laya also trains its larger native decision head. Counts for linear classifiers are not recorded in this column. Original weights are cached pretrained input files, not packed deployment size. Saved checkpoints may include optimizer/RNG state. Peak VRAM is torch-allocated memory, excluding driver/display overhead; the latency table reports inference peak, with training peak listed separately.

Model	Feature documents	Recorded compute s	Feature peak GiB	Cache reused
FinBERT logistic	5200	157.1	0.51	5200
BGE-small logistic	5200	85.7	0.18	5200

Frozen-feature counts include train, selection and calibration. Recorded compute sums each requested document's original preprocessing/encoder time under the pinned recipe; it excludes model loading and cache I/O. Cached rows retain this measurement when reused across sample sizes. Fit elapsed includes only

preparation performed during that fit. These timings describe the shared machine, not a new cold run.

Critical-error inspection

Routine probability 0.993; 2026-07-01; releases; source. Teacher label: review_worthy. Body hash: 0d3834cf8f28a6d0739cad71e75a30257ccda1dfb5ed267e15f02aeb30d7b2e9.

Exhibit 99.1 SunPower Closes $10 Million Share Exchange OREM, Utah (July 1, 2026) – SunPower Inc. (“SunPower,” the “Company,” or Nasdaq: “SPWR”), a solar technology, ...

Routine probability 0.992; 2026-07-09; fed; source. Teacher label: review_worthy. Body hash: 1b719fccf4a881f128158f176db03990a19fb1a4691b437b11575a6f50262801.

July 09, 2026 Federal Reserve announces the leadership and objectives of its task forces to advance the conduct of monetary policy For release at 3:00 ...

Routine probability 0.991; 2026-07-28; releases; source. Teacher label: review_worthy. Body hash: 6dc565847109e454abd0548bed4c106ecae9928e51eb97fc0bd3ac47b2e2f3ad.

Exhibit 99.1 CARPENTER TECHNOLOGY ANNOUNCES THE PASSING OF BRIAN MALLOY TONY THENE TO RETURN TO CEO ROLE EFFECTIVE IMMEDIATELY PHILADELPHIA, July 27, 2026 (GLOBE NEWSWIRE) ...

Routine probability 0.988; 2026-09-02; releases; source. Teacher label: review_worthy. Body hash: 3addec01a59c450f3327de57f7457c0437d24e394997eb53d7429142e14e8547.

Exhibit 99.1 FINANCIAL CONTACT: MEDIA CONTACT: Anoori Naughton Allison Kleinfelter anaughton@hersheys.com akleinfelter@hersheys.com The Hershey Company Names Dave Hulays Chief Financial Officer Hulays Succeeds Steve Voskuil, ...

Routine probability 0.985; 2026-09-08; releases; source. Teacher label: review_worthy. Body hash: 5f0d2e0dae7a770464b15824dfc772963cb6633a7bc57586cd35244f85cd7823.

Exhibit 99.1 Shutterstock Appoints Three Accomplished Business Leaders to Board of Directors Timothy Adams, Matthew Salzberg, and Michael Thompson bring extensive financial, operational, investment and ...

Routine probability 0.976; 2026-09-18; releases; source. Teacher label: review_worthy. Body hash: 0eadd9c1aa56af64b24bbb86d986ed263ea5f3ae7a7522794a53ccb22b745e52.

Exhibit 99.3 FOR IMMEDIATE RELEASE September 18, 2026 Copper Property CTL Pass Through Trust Releases Q2-2026 Penney Intermediate Holdings LLC Financial Statements and Master Lease ...

Routine probability 0.973; 2026-08-24; releases; source. Teacher label: review_worthy. Body hash: 2f7e803ea19f2ebabd62406578ce98b2ced39f4244ed9659d1c88bfc24842374.

Exhibit 99.2 Ocean Power Technologies, Inc. Provides Required Disclosure MONROE TOWNSHIP, N.J., Aug. 21, 2026 (GLOBE NEWSWIRE) — Ocean Power Technologies, Inc. (“OPT” or the ...

Routine probability 0.970; 2026-07-07; releases; source. Teacher label: review_worthy. Body hash: 444123abdd397d23d43ae9b70035d42fdb85592791182721b2b3461e64401e62.

Exhibit 99.1 Zapata Quantum to Present at the Global Technology Virtual Investor Conference July 9th CEO Sumit Kapur will discuss highlights including the Company's NVIDIA ...

Error examples are selected after testing for inspection, not model tuning. Labels may be wrong; analyst usefulness needs independent review. All miss identities and model disagreements remain inspectable in results.json.

Reproducibility and limits

The serial runner stores failed/OOM configurations, any window-batch fallback, selected recipes, pinned model revisions, library versions, source snapshots, RNG/optimizer checkpoints, dataset/subset hashes and an immutable pre-test freeze. Reproduce with the locked dependencies and study.py --stage all; existing runs resume with changed-artifact guards. Raw-source rights/attribution travel with the corpus. Old API/container behavior remains separate.

This is source-only inbox triage on constrained public cohorts, not expert materiality assessment or a trading strategy. One seed cannot quantify training instability. Bootstrap intervals quantify cohort sampling only. Teacher labels favor agreement with the labeling approach; independent audit availability is stated above. Missing source/date coverage, approximate duplicate retrieval and pretraining exposure restrict generalization. No paid SOTA comparison or RL/RLCD result is claimed. RL/RLCD remains a possible follow-on after the core study.

Exploratory diagnostics added after the primary results

The following questions were added after inspecting the primary held-out results. They are descriptive, not prespecified hypotheses, new model choices or tuned discard rules. The original primary comparison and its conclusion remain unchanged. Intervals condition on one trained seed and have no multiplicity adjustment.

Approximately 68.3% of the observed macro-F1 point difference comes from the unclear class. Each class contributes its F1 difference divided by three; source-family F1 scores cannot be added this way. The contribution intervals all include zero. This decomposition is not a causal attribution or a resolved class-specific gain.

Class	Class F1 difference	Contribution to macro-F1 difference	Contribution 95% CI
routine	0.0051	0.0017	[-0.010, 0.013]
review_worthy	0.0053	0.0018	[-0.004, 0.007]
unclear	0.0225	0.0075	[-0.008, 0.022]

Both review-worthy and unclear predictions route to an analyst. The derived two-class view below collapses those existing actions, without changing a prediction. Its numerical scale and meaning differ from the original three-class endpoint; it does not replace it.

Model	Routing macro F1	95% CI	Review misses	Review support	Unclear discarded	Workload removed
Laya LoRA	0.773	[0.748, 0.796]	191	1499	14	0.234
Word TF-IDF logistic	0.768	[0.742, 0.791]	212	1499	25	0.249

Paired routing-F1 difference interval: [-0.018, 0.028]. Laya routes 28 additional documents for review in this cohort. Its lower point count of required-review misses therefore comes with additional workload; neither a routing-F1 nor a miss-rate advantage is resolved by these intervals.

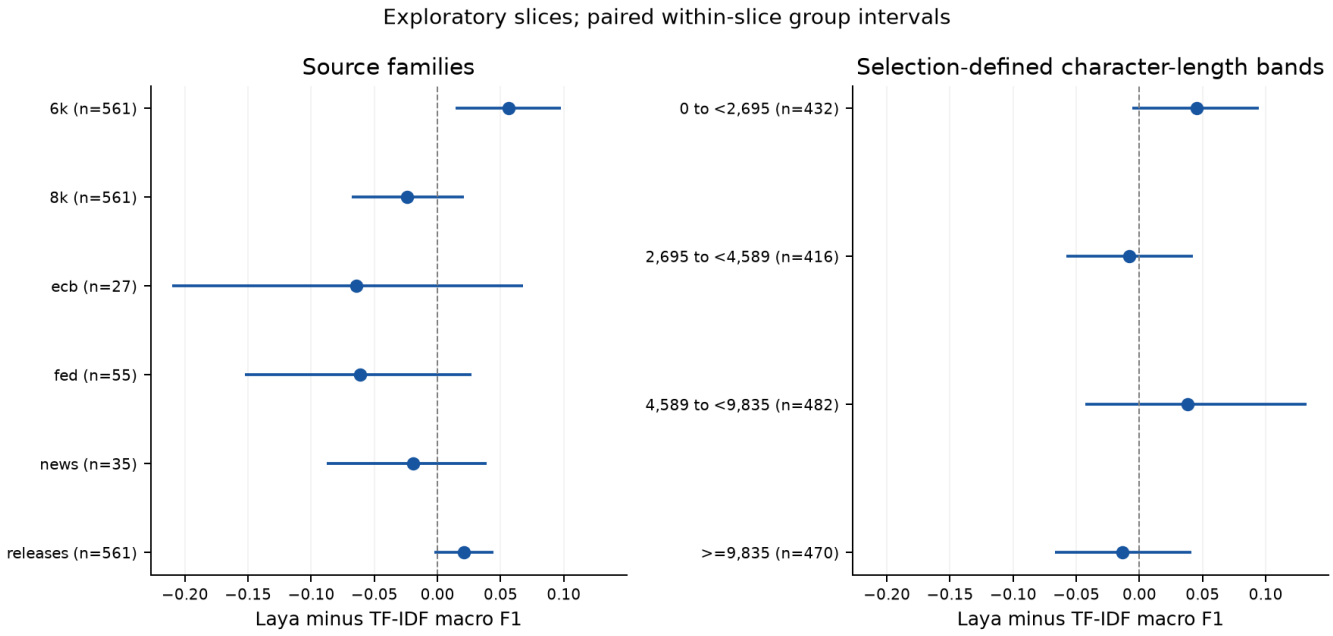
Shared and model-specific critical misses

Critical-miss pattern	Documents
Both models	105
Laya only	72
TF-IDF only	82
Neither model	1033

These four cells partition 1292 teacher review-worthy documents. Overall three-class prediction agreement is 79.56%. Complementary errors motivate a future ensemble/cascade study, but these counts omit its

false-positive, workload and compute tradeoffs. They do not measure a new deployed policy. Any such policy needs validation selection and fresh test data. Common misses cannot distinguish student errors from questionable teacher references without independent review.

Source and document-length differences



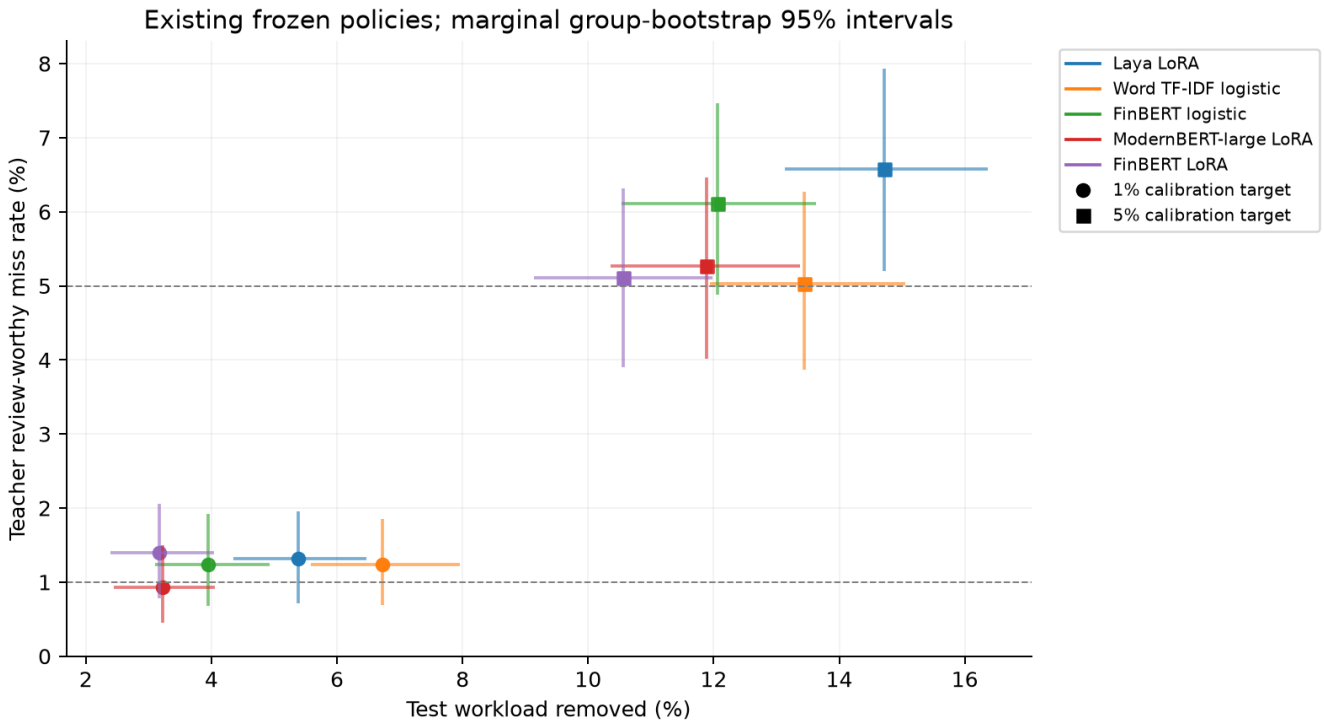
Primary-model differences within source and length slices

Dots are Laya minus TF-IDF point differences; bars are paired 95% intervals from 2,000 whole-group resamples within each slice. Length bands use character-count quartiles from the 837-document selection cohort, chosen mechanically for this post-hoc analysis. Their cutpoints are 2,695, 4,589, 9,835 characters. Test bands need not have equal counts. Character lengths are not model token counts. Source and length overlap, so these slices do not isolate a causal effect of context or architecture.

Family	Documents	Routine	Review-worthy	Unclear	Fixed-label F1 ceiling
6k	561	105	297	159	1.000
8k	561	100	413	48	1.000
ecb	27	9	18	0	0.667
fed	55	15	40	0	0.667
news	35	10	25	0	0.667
releases	561	62	499	0	0.667

All slice scores retain the three declared labels. A slice without unclear references has a maximum possible macro F1 of 2/3 under this definition, not one. Compare models within the same slice; raw scores across families have different label support and attainable ceilings. The 6-K slice has the largest positive point difference; its nominal interval excludes zero. This exploratory, uncorrected comparison does not establish universal source superiority. All four length-band intervals include zero, including the longest band.

Uncertainty around the frozen discard rules



Workload and critical misses for the existing frozen policies

Circles/squares identify the original 1%/5% calibration targets. Dashed lines show those targets, not deployment guarantees. Horizontal and vertical bars are separate marginal 95% group intervals, not joint confidence regions. These points and intervals reuse the original frozen policies and statistics; no threshold sweep, ensemble or model selection was performed.

Calibration target	Workload-difference 95% CI (pp)	Critical-miss-rate difference 95% CI (pp)
0.01	[-2.50, -0.22]	[-0.61, 0.78]
0.05	[-0.22, 2.67]	[0.08, 2.90]

Differences are Laya minus TF-IDF. More workload removed is favorable; a higher critical-miss rate is unfavorable. At the 1% target, Laya removes less workload and the miss-rate difference is unresolved. At the 5% target, its nominal interval indicates more teacher review-worthy misses while the workload difference remains unresolved. These conditional, uncorrected comparisons help interpret the existing systems; they are not permission to choose a new operating rule from this test set.